A Concrete Relative Projection Constant Equal to 7/3

Run `fa_banach_001`, lane `agent_lane_01`

2026-06-14

Source Problem

This packet addresses the final question in Giuliano Basso's *Computation of maximal projection constants*, arXiv:1901.07866 [1]. In Remark 4.6, after listing

$$\Pi(1,1) = 1, \qquad \Pi(2,3) = \frac{4}{3}, \qquad \Pi(4,6) = \frac{5}{3}, \qquad \Pi(5,10) = 2,$$

the source asks:

$$\text{Are there integers } n \leq d \text{ such that } \Pi(n,d) = \frac{7}{3}?$$

Here

$$\Pi(n,d) = \sup\{\Pi(E) : E \subset \ell_\infty^d \text{ is an } n\text{-dimensional Banach space}\}.$$

as claimed.

Remark 4.6 As $\mathbb{R}^5$ admits a system of ten equiangular lines, cf. [LS73, p. 496], the general upper bound of König, Lewis, and Lin, cf. [KLL83], tells us that

$$\Pi(5,10) = 2.$$

To summarize, we obtain the sequence

$$\Pi(1,1) = 1, \quad \Pi(2,3) = \frac{4}{3}, \quad \Pi(4,6) = \frac{5}{3}, \quad \Pi(5,10) = 2,$$

which naturally leads to the open question: Are there integers $n \leqslant d$ such that

$$\Pi(n,d) = \frac{7}{3}?$$

4.5 Acknowledgements: I am thankful to Urs Lang who read earlier draft versions of this article and who stimulated several improvements. Moreover, I am grateful to Anna Bot for proofreading this paper.

Figure 1: Basso, Remark 4.6, where the 7/3 question is posed.

Background Used

The same source recalls the theorem of Koenig–Lewis–Lin [2]:

$$\Pi(n, d) \leq \frac{n}{d} + \sqrt{(d-1)\frac{n}{d}\left(1 - \frac{n}{d}\right)},$$

with equality if and only if $\mathbb{R}^n$ admits a system of d distinct equiangular lines.

Theorem 1.2 *If* $n \geqslant 1$ *is an integer, then*

$$\Pi_n = \sup_{d \geqslant 1} \max \left\{ \frac{n}{d} + \frac{1}{d} \sum_{k=1}^{n} \lambda_k(T) \ : \ T \textit{ is a } K_{n+2}\textit{-free two-graph of order } d \right\}.$$

To prove Theorem 1.2, we invoke a simple trick, see Lemma 2.2, that allows us to greatly narrow down the matrices that need to be considered. This is done in Section 2.

1.4 Relative projection constants The following question has first been systematically addressed by König, Lewis, and Lin in [KLL83]:

Question 1.1 *Let* $n, d \geqslant 0$ *be integers. What is*

$$\Pi(n, d) := \sup \left\{ \Pi(E) : E \subset \ell_d^\infty \textit{ is an } n\textit{-dimensional Banach space} \right\}?$$

By definition, $\sup \varnothing = -\infty$. Clearly, $\Pi(d, d) = 1$ and it is a direct consequence of the classical Hahn-Banach theorem that $\Pi(1, d) = 1$ for all integers $d \geqslant 1$. The quantity $\Pi(d-1, d)$ has been examined by Bohnenblust, cf. [Boh38], where it is shown that $\Pi(d-1, d) \leqslant 2 - \frac{2}{d}$. In [CL09], Chalmers and Lewicki determined the exact value of $\Pi(3, 5)$. In [KLL83], König, Lewis, and Lin established the general upper bound

$$\Pi(n, d) \leqslant \frac{n}{d} + \sqrt{(d-1)\frac{n}{d}(1 - \frac{n}{d})}$$

with equality if and only if $\mathbb{R}^n$ admits a system of d distinct equiangular lines. Thereby, as $\mathbb{R}^3$ admits a system of six equiangular lines, cf. [LS73, p. 496], it holds that

$$\Pi(3, 6) = \frac{1 + \sqrt{5}}{2}.$$

In light of

$$\Pi(4, 6) = \frac{5}{3},$$

which we demonstrate in Paragraph 4.4, up to $d = 6$ all exact values of $\Pi(n, d)$ for $1 \leqslant n \leqslant d$ are now computed. It is well-known that

Figure 2: Basso's definition of $\Pi(n, d)$ and the quoted Koenig–Lewis–Lin equality criterion.

Proof Intuition

The value 7/3 is forced by the Koenig–Lewis–Lin bound when $(n, d) = (7, 15)$:

$$\frac{7}{15} + \sqrt{14 \cdot \frac{7}{15} \cdot \frac{8}{15}} = \frac{7}{15} + \frac{28}{15} = \frac{7}{3}.$$

So the problem reduces to a finite-dimensional geometric certificate: produce 15 equiangular lines in a 7-dimensional real space. The standard 28-line configuration in $\mathbb{R}^7$ gives this immediately, and the explicit model below keeps the reduction checkable.

The Answer

Theorem 1.

$$\Pi(7,15) = \frac{7}{3}.$$

Proof. Let $H = \{x \in \mathbb{R}^8 : \sum_{k=1}^{8} x_k = 0\}$. Then H is a 7-dimensional real Hilbert space. For each two-element set $A = \{i,j\} \subset \{1,\dots,8\}$, define

$$v_A = e_i + e_j - \frac{1}{4}(1,1,1,1,1,1,1,1) \in H.$$

For two such sets A and B,

$$\langle v_A, v_B \rangle = |A \cap B| - \frac{1}{2}.$$

In particular, $\|v_A\|^2 = 3/2$, and for $A \neq B$ the absolute value of $\langle v_A, v_B \rangle$ is $1/2$. Thus the lines spanned by the nonzero vectors v_A are equiangular, with common normalized absolute inner product $(1/2)/(3/2) = 1/3$.

Select the following 15 two-element sets:

$$\{1,j\}\ (2 \le j \le 8), \qquad \{2,j\}\ (3 \le j \le 8), \qquad \{3,4\}, \{3,5\}.$$

The first seven selected vectors $v_{\{1,j\}}$, $2 \le j \le 8$, already span H. Indeed, if

$$\sum_{j=2}^{8} a_j v_{\{1,j\}} = 0$$

and $A = \sum_{j=2}^{8} a_j$, then the k-th coordinate for $2 \le k \le 8$ gives $a_k - A/4 = 0$. Hence all $a_k = A/4$, so

$$A = 7A/4,$$

which forces $A = 0$ and therefore $a_k = 0$ for all k. Since H has dimension 7, these seven vectors are a basis of H.

Consequently $H \cong \mathbb{R}^7$ admits 15 distinct equiangular lines. Applying the Koenig–Lewis–Lin equality criterion quoted above,

$$\Pi(7,15) = \frac{7}{15} + \sqrt{14 \cdot \frac{7}{15} \cdot \frac{8}{15}} = \frac{7}{15} + \frac{28}{15} = \frac{7}{3}.$$

□

Verification Notes

The script `code/verify_equiangular_7_15.py` checks the construction with exact rational arithmetic. It verifies that the 15 selected vectors have rank 7, squared norm $3/2$, off-diagonal absolute inner product $1/2$, and that the Koenig–Lewis–Lin value for $(n,d) = (7,15)$ is exactly $7/3$.

Limitations and Novelty Check

This is a short finite-dimensional observation. It relies on the Koenig–Lewis–Lin equality theorem as quoted in the source paper, and on the explicit equiangular-line configuration written above. Cheap run-index searches found no existing packet or attempt for arXiv:1901.07866 or the phrase $\Pi(n,d) = 7/3$. Bounded web searches on 2026-06-14 for exact phrases around $\Pi(n,d)$, 7/3, projection constants, and equiangular lines returned no exact answer hit.

Human Review Recommendation

Review the application of the Koenig–Lewis–Lin equality criterion and the rank/equiangular-line calculation. The latter is elementary and independently checked by the verifier script.

References

[1] G. Basso, *Computation of maximal projection constants*, arXiv:1901.07866, 2019.

[2] H. Koenig, D. R. Lewis, and P.-K. Lin, *Finite dimensional projection constants*, Studia Math. 75 (1983), 341–358.

[3] P. W. H. Lemmens and J. J. Seidel, *Equiangular lines*, J. Algebra 24 (1973), 494–512.